

Haoyu Wang

✉ HAW309@pitt.edu

🌐 <https://carlwhy-28.github.io>

🎓 Google Scholar

Education

University of Pittsburgh, School of Medicine

Pittsburgh, PA

Ph.D. in Biomedical Informatics and Artificial Intelligence

Aug 2025 – 2029 (Expected)

- Focus: Computational Biology, Machine Learning, Spatial Transcriptomics

Beijing Forestry University

Beijing, China

B.E. in Computer Science and Technology

Sep 2020 – Jun 2024

Work Experience

University of Pittsburgh Medical Center

Computational Biology Intern

Aug 2024 – Jul 2025

- Designed and implemented dual-graph attention neural networks (DGAT) for spatial multi-omics integration, enabling protein expression imputation from transcriptomic and spatial features
- Conducted downstream pathway and transcription factor activity analysis using statistical modeling and enrichment methods to identify regulatory signals in tumor microenvironments

Selected Research Experience

DGAT: Spatial Protein Landscapes Imputation from Transcriptomics

Primary Author; Computational Framework & Model Development

- Designed and developed a spatially-aware GAT-based model for protein imputation from mRNA and spatial data
- Improved protein-wise correlation by 5%–20% over 7 prior methods in held-out sample benchmarking
- Achieved 1%–3% gains in Adjusted Rand Index for spatial domain clustering

DGATviz: Web Platform for Spatial Protein Imputation

Lead Developer; Full-Stack Development

- Developed a web platform for DGAT inference, visualization, and result download
- Built an asynchronous pipeline using Streamlit, MySQL, object storage, and Python workers
- Integrated task tracking, email notifications, and pickup-code-based result retrieval

STAN: a Computational Framework Inferring Transcription Factor Activity

Image Feature Extraction and Downstream Analysis

- Applied unsupervised deep learning methods for whole-slide image feature extraction to support downstream modeling
- Performed downstream analysis of spatial transcriptomics data to infer transcription factor and pathway activities

Publications

Wang, H., Batz, C., Osmanbeyoglu, H.U., DGAT: A Dual-Graph Attention Network for Inferring Spatial Protein Landscapes from Transcriptomics. *bioRxiv* [Preprint], 2025 Jul 9. doi: [10.1101/2025.07.05.662121](https://doi.org/10.1101/2025.07.05.662121)

Published in **Nature Communications**.

Abstract selected for oral presentation at ISMB 2025; workshop presentation at MLCB 2025; tutorial at ECCB 2026.

Linan Zhang, April Sagan, Bin Qin, Haoyu Wang, Elena Kim, Baoli Hu, Hatice Ulku Osmanbeyoglu, STAN, a computational framework for inferring spatially informed transcription factor activity, *Nucleic Acids Research*, Volume 54, Issue 1, 13 January 2026, gkaf1473. doi: [10.1093/nar/gkaf1473](https://doi.org/10.1093/nar/gkaf1473)

Technical Skills

- **Programming & Tools:** Python, SQL, R, Java, C++, MATLAB, Linux (Ubuntu), Git, SLURM
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, Graph Neural Networks, Transformers, Computer Vision
- **Bioinformatics:** Spatial Transcriptomics, Single-cell Analysis, Genomics, Pathway Modeling